LaRS Embedding Distribution Analysis

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1. Introduction

This report analyses how the LaRS (Large Real-world Shoreline) dataset's test set is distributed relative to the training set in the embedding space produced by NVIDIA RADIO-H, a large vision foundation model trained to distil CLIP, DINOv2, and SAM into a single encoder. The goal is to identify potential out-of-distribution (OOD) test images, understand dataset shift, and estimate where a model trained on the LaRS training set is most likely to fail.

The analysis covers three aspects:

1. **t-SNE visualisation** — global structure of the embedding space, coloured by split and by sequence.
2. **Cosine similarity distribution** — per-image quantitative coverage score: how similar is each test image to its nearest training neighbour?
3. **Sequence coverage** — whether the physical environments (camera sequences) in the test set were seen during training.

2. Dataset Overview

Split	Images	Sequences	Resolution
Train	2,605	1,792	1278×958 / 1280×720
Val	198	77	same
Test	1,203	440	same

> **Key structural finding:** The test set contains **440 sequences**, none of which appear in the training set (0 overlap). Every test sequence is from a physical environment or recording session completely unseen during training. This is by design for a rigorous benchmark, but it means the model must generalise

across camera locations and scene types, not just across time within the same scene.

Note on database schema: The original analysis request asked for per-patch fields (`patch_id`, `patch_start`, `patch_end`) in the SQLite database. These fields are **not present** in the database because the approach was changed to use RADIO's internal spatial tokens rather than explicit patch extraction (see Section 3.1). Instead of storing one row per crop, the database stores one row per image with a single mean embedding vector. The field `img_set` requested is stored as `split`. If per-crop embeddings with position metadata are required, the pipeline would need to be modified to extract explicit crops and store their coordinates.

3. Methodology

3.1 Embedding model — RADIO-H with internal spatial tokens

Each image is passed through **RADIO-H** (nvidia/C-RADIOv3-H, 653 M parameters) at its original resolution. RADIO uses a Vision Transformer (ViT) with a patch size of 16 px, meaning the image is divided into a grid of 16×16 px cells; a 1280×960 image produces 80×60 = 4,800 spatial tokens. Each token is a 1280-dimensional vector produced by the transformer attention mechanism — crucially, every token has attended to all other tokens in the image, so each embedding captures its local region in full global context. Unlike manually cropping images into patches and running each crop separately (which produces context-free, isolated embeddings), this approach requires only one forward pass per image and produces richer representations.

The spatial tokens are averaged across the spatial dimension to produce a single 1280-dimensional mean vector per image (`spatial_mean_embedding`). This is used for all analysis in this report. RADIO also produces a global summary token (`summary_embedding`, 3840-dimensional) analogous to the [CLS] token in ViT — this is stored in the database but not used here, because the spatial mean better represents the distribution of local features across the image, which is more appropriate for detecting regional distributional shift than a single global descriptor. Both are available in the SQLite database for future use.

3.2 Dimensionality reduction

Raw 1280-dimensional spatial mean embeddings are first reduced to 50 dimensions with PCA — retaining **84.3% of the total variance** — then projected to 2D with t-SNE (GPU-accelerated via cuML, RAPIDS 24.12). PCA pre-reduction is standard practice before t-SNE: it removes high-frequency noise while preserving dominant structure and dramatically speeds up the t-SNE computation. The final t-SNE achieved a **KL divergence of 0.239**, indicating a good-quality 2D embedding (lower is better; values below ~1.0 are generally considered reliable for datasets of this size).

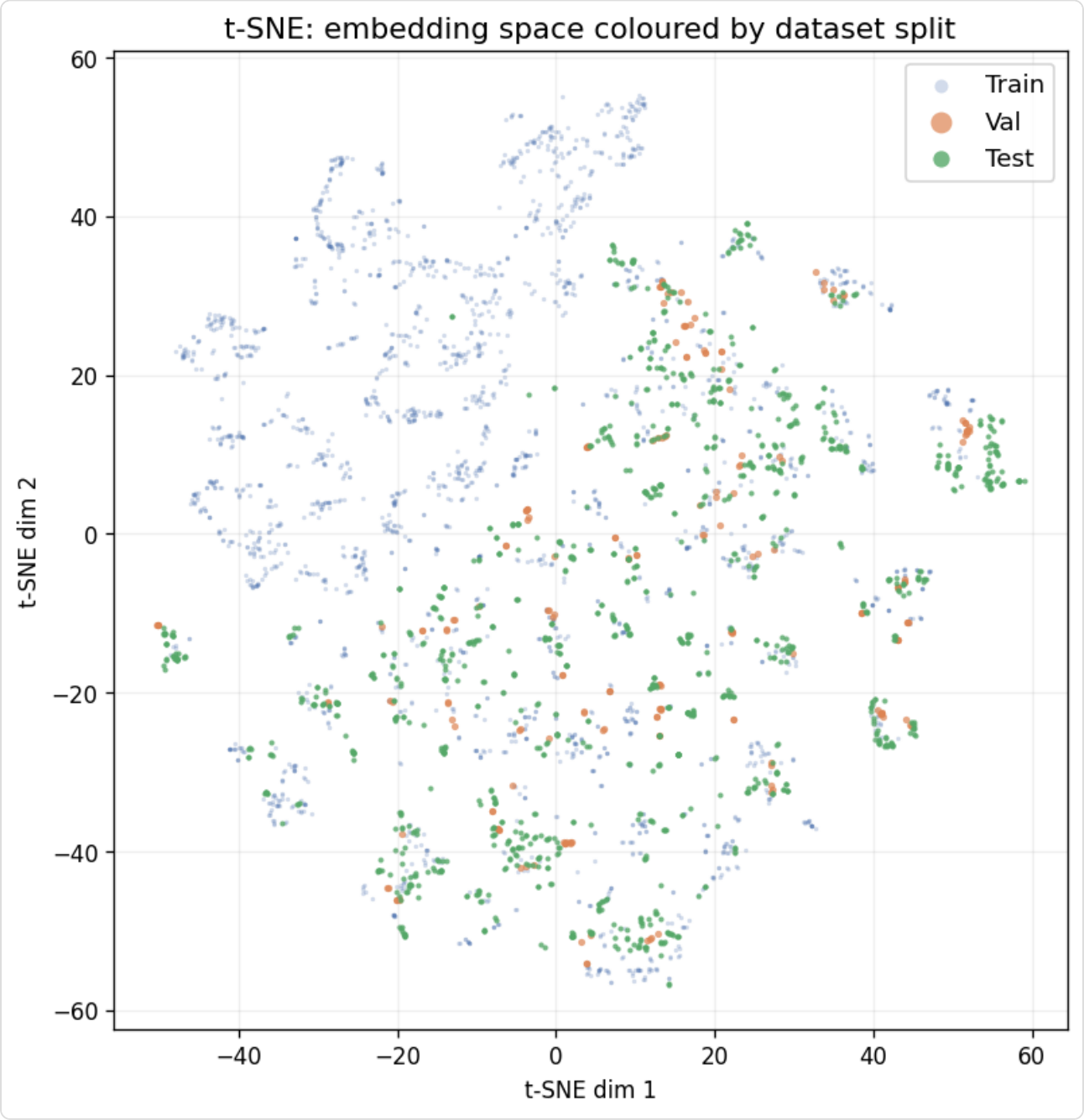
3.3 Cosine similarity coverage score

For each non-training image, the maximum cosine similarity to any training image is computed as a scalar coverage score in [0, 1]. A score near 1.0 means the image is visually very similar to at least one training image; a lower score indicates the image is in a region of the embedding space less covered by training. The table in Section 5 reports coverage at multiple thresholds (0.85, 0.80, 0.75, 0.70) so the reader can assess sensitivity — the choice of threshold significantly affects what fraction of images are classified as at-risk.

4. t-SNE Visualisation

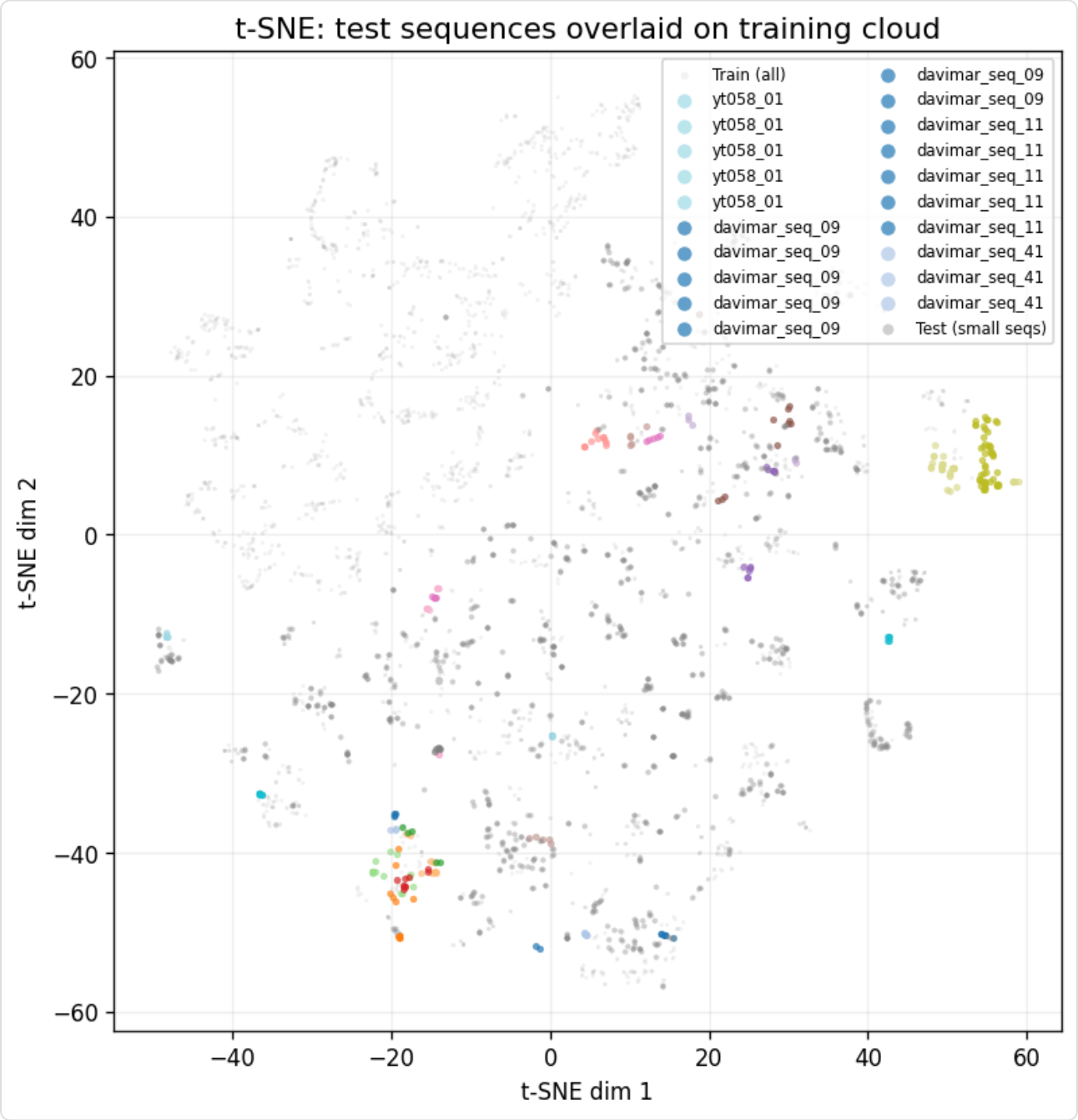
Note: the figures below are static (matplotlib). Interactive versions with per-point image hover are available in `analysis.ipynb` — open it in Jupyter to explore individual images within any cluster.

Figure 1 — Split distribution



Train, val, and test images largely overlap in embedding space, which is consistent with the high cosine similarity scores (see Section 5). The test set does not form isolated clusters — it is interspersed throughout the training cloud. This suggests that the visual content of test scenes (water surface, obstacles, lighting conditions) is broadly similar to what the model sees during training, even though the specific camera environments are all new.

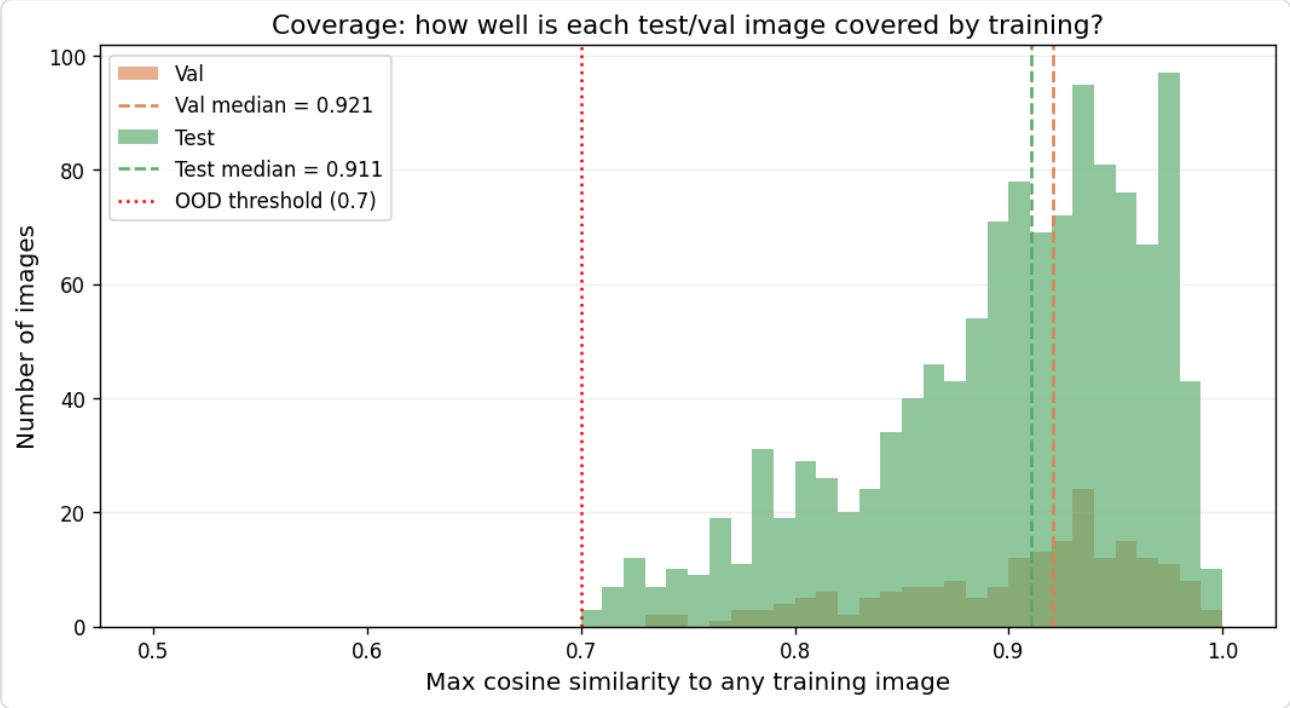
Figure 2 — Test sequence overlay



Colouring test images by their sequence ID reveals whether certain environments cluster in a specific region of embedding space (potential domain shift). Sequences that appear as isolated islands away from the grey training cloud are the highest-risk evaluation environments — the model has seen nothing visually similar during training.

5. Cosine Similarity Coverage Analysis

Figure 3 — Coverage histogram



The table below reports the percentage of images below several thresholds. No single threshold is objectively "correct" — the choice depends on how conservative you want to be when identifying at-risk images. The 0.85 threshold is a stricter definition of good coverage; 0.70 is a very conservative floor.

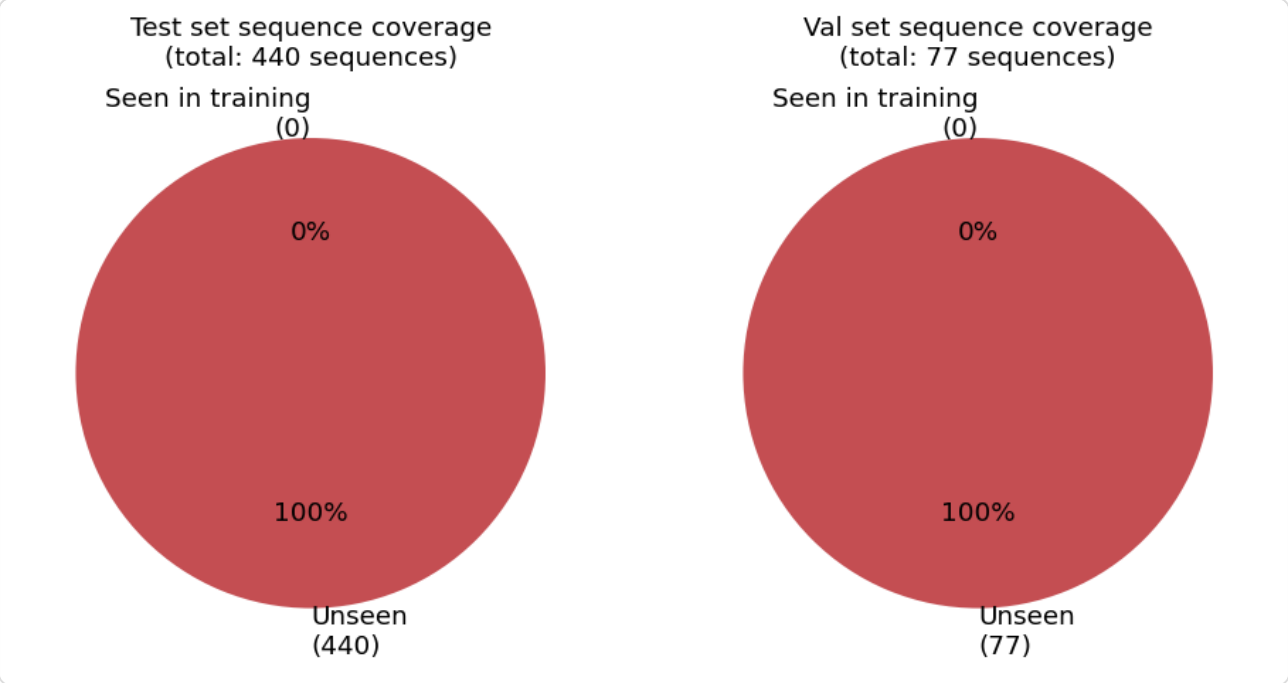
Split	Images	Median max-sim	Below 0.85	Below 0.80	Below 0.75	Below 0.70
Val	198	0.921	19.7% (39)	7.6% (15)	2.0% (4)	0.0% (0)
Test	**1,203**	**0.911**	**21.7% (261)**	**10.6% (128)**	**3.2% (39)**	**0.0% (0)**

Interpretation: Despite every test sequence being physically unseen during training, the cosine similarity distribution is concentrated above 0.9 for most images. This indicates that the visual appearance of test images (sea surface texture, lighting, obstacle types) is broadly represented in the training set, even if the exact locations are not. However, note that high similarity in RADIO's general-purpose embedding space does not guarantee that a task-

> specific segmentation model will not fail — it measures visual distributional overlap, not segmentation difficulty. The images in the lowest-similarity tail (below 0.80, roughly 10.6% (128) of test images) are the strongest candidates for targeted data augmentation.

6. Sequence Coverage

Figure 4 — Sequence coverage by split



The left pie (test set) confirms that 100% of test sequences are physically absent from training — all 440 test sequences are from environments the model has never seen. The right pie (val set) shows the same pattern: all val sequences are also unseen. This is consistent with a benchmark designed to measure generalisation rather than memorisation. It means no sequence-level adaptation (e.g. test-time fine-tuning per camera) is possible — models must rely entirely on what they learned from the training sequences. Combined with the cosine similarity findings in Section 5, the picture is: the visual content types generalise well (high similarity scores), but the specific environments are all new.

7. Most At-Risk Test Images (Predicted OOD)

The following test images have the lowest maximum cosine similarity to any training image, making them the highest-risk candidates for model failure. **Important caveat: no segmentation model has been evaluated at this stage.** These are predicted at-risk images based on embedding distance — images in regions of visual space underrepresented in training. Actual failure cases can only be confirmed by running inference and inspecting predictions.

#	Thumbnail	Image name	Sequence	Max cosine sim ↑
1		`yt017_00_00049`	yt017_00	**0.7040**
2		`inhouse_seq_111_01550`	inhouse_seq_111	**0.7060**
3		`inhouse_seq_111_00642`	inhouse_seq_111	**0.7099**
4		`yt017_01_00149`	yt017_01	**0.7134**
5		`yt049_01_00080`	yt049_01	**0.7166**
6		`yt049_02_00180`	yt049_02	**0.7168**
7		`yt066_05_00060`	yt066_05	**0.7169**

				
8		yt097_00_00419	yt097_00	**0.7183**
9		yt097_00_00149	yt097_00	**0.7184**
10		inhouse_seq_111_01729	inhouse_seq_111	**0.7192**
11		yt017_00_00154	yt017_00	**0.7202**
12		yt097_00_00049	yt097_00	**0.7208**
13		yt017_01_00059	yt017_01	**0.7230**
14		yt049_02_00080	yt049_02	**0.7233**
15		yt066_04_00060	yt066_04	**0.7242**

> **Note:** Even the most at-risk images have max-sim above 0.70. At a 0.80 threshold, roughly 10.6% (128) of test images are flagged — these are the strongest candidates for closer inspection and targeted data augmentation. The at-risk images tend to cluster in a small number of sequences (e.g. `yt017`), suggesting a few specific environments drive most of the distributional tail rather than a broad spread of OOD content.

8. Conclusions

- The LaRS test set is **visually broadly covered** by the training set despite containing entirely new camera sequences. Median test image max cosine similarity to training is 0.911, suggesting most test images are visually similar to at least one training image.
- The t-SNE distributions of train and test overlap substantially, indicating **no large-scale domain gap** in general visual features. However, this is measured in a general-purpose embedding space (RADIO-H) — it does not guarantee that a task-specific segmentation model will perform well, as segmentation difficulty depends on factors (obstacle size, label ambiguity, edge cases) not fully captured by embedding distance alone.
- Roughly **10.6% (128) of test images fall below a 0.80 cosine similarity threshold**. These are the priority candidates for targeted data augmentation or additional annotation before training.
- Since all test and val sequences are physically disjoint from training, model performance will depend on generalisation of visual patterns (water texture, obstacle appearance, lighting), not memorisation of specific scene geometry.
- The at-risk images are concentrated in a small number of sequences rather than spread uniformly — a few specific environments (e.g. `yt017`) drive most of the distributional tail and deserve targeted attention.
